

Sneh Omer

Third Year Undergraduate
Department of Chemical Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech.	Indian Institute of Technology Kanpur	7.8/10
2020	Class XII (CBSE)	Delhi Public School, Kanpur	97.0 %
2018	Class X (CBSE)	UP Kirana Seva Samiti Balika Vidyalaya, Kanpur	95.6 %

Scholastic Achievements

- Selected as one of the top **400** students from **1600+** applicants for the prestigious **SURGE** summer internship program
- Secured **All India Rank 14722** in **Joint Entrance Exam Advanced 2021** among the **1.5 Lakh** shortlisted candidates
- Secured **All India Rank 8483** in **Joint Entrance Exam Main 2021** among the **12 Lakh** registered applicants

Work Experience

Ink Optimization for Tactile Reusable and Reprintable Paper | *SURGE Internship* (May'23 - Jul'23)
Mentor: Prof. Animangsu Ghatak, Department of Chemical Engineering

Objective	● Analysing Ink Formulation and Properties for Enhanced Reusable and Reprintable Paper Technology
Approach	● Generated temperature-dependent height-versus-time graphs for ink dots, giving valuable insights of ink performance ● Utilized the Owens-Wendt-Rabel-Kaelble method to compare the surface energies for compatibility assessments ● Successfully attained a defined threshold-force for the required detachment of ink by utilizing Young's Equation
Impact	● Devised a novel strategy to write tactile text on reusable paper which led to poster presentation at IIT Kanpur

Key Projects

- Deep Learning in Chemical Engineering** | **SimuTech, Chemineers - IITK** | 🌐 (Feb'23 - Jun'23)
 - Developed an **ANN** to accurately predict **adsorption capacity**, enhancing modeling capabilities for adsorption processes
 - Designed a **CNN model** on **COMSOL simulated images** to distinguish between **turbulent and laminar flows**
 - Deployed and compared multiple CNN models like **ResNet, VGG16, MobileNet, LeNet** with highest accuracy of **0.90**
- Computational analysis of T vs x, y diagram of binary solution** | **Self Project** | 🌐 (Dec'22 - Jan'23)
 - Visualised the experimentally determined T-x,y and x-y data of the binary solution of ethyl acetate and 2-ethoxy ethanol
 - Utilised **Modified Rault's Law** to analyze the experimental data of mole fraction of ethyl acetate in liquid and gas phase
 - Forged an algorithm on MATLAB to compare experimental T-x,y data with the theoretical **Margules Activity Model** data
- Virtual Chemical Company** | **CHE261A Course Project** | **Prof. Raghvendra Ragipani** (Dec'22 - Jan'23)
 - Contributed to the success of **Synthochem**, specializing in high-value chemical production from **fine & speciality chemicals**
 - Successfully designed and developed a **scalable process** for a plant with a production capacity of **1000 kg/day** of product
 - Conducted **material balance, unit operation identification** for optimal condition and capital **cost determination**

Technical Skills

- Programming Languages & Libraries:** C, C++, Python, MATLAB, Pandas, NumPy, Matplotlib, TensorFlow
- Software & Utilities:** Aspen, COMSOL, Simulink, VS Code, SQL, L^AT_EX, Jupyter Notebook, Fusion360

Relevant Course Work

Heat Transfer and its Application	Thermodynamics	Chemical Engineering Thermodynamics
Chemical Process Simulation Lab	Mass Transfer and its Application	Chemical Process Industries
Computational Methods in Engineering	Intro to CHE & Process Calculation	Engineering Graphics
Linear Algebra and ODE	Introduction to Electronics	Introduction to Economics

Positions of Responsibility

- Manager & Mentor** | **Chemineers Society, IIT Kanpur** (Jun'23 - Present)
 - Managed the selection process of **15+ secretaries** in order to form a well-organised **3-tier team** of **30+ members**
 - Spearheading **Events & Design team** of **7 secretaries** to boost the reach of Chemineers Society among IITK community
 - Mentored a **semester project** for second year students, Analyzing Heat Transfer: from theory to practical application
 - Organized **academic and research-oriented sessions** e.g. Intern Session, Alumni Talks and Open House with professors
- Maintenance Secretary** | **Hall of Residence VI, IIT Kanpur** (May'23 - Present)
 - Developed comprehensive maintenance strategy in collaboration with hostel authorities, optimizing upkeep of hostel facilities
 - Streamlined **resource allocation** by assisting in **expense budgeting**, facilitating procurement of essential hostel assets
 - Efficiently prioritized the maintenance requests, ensuring **prompt issues resolution** and seamless coordination with staff

Extra-Curricular Activities

- Mentored **5** first-year students, aiding their smooth transition and contributed in Orientation Program 2022 as **Student Guide**
- Secured **4th position** in the inter-school **WIZ National Spell Bee Competition** and qualified for the State level competition
- Ranked among the **top 50 participants** in the **National Science Olympiad**, competing against 50+ schools from 5 cities